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Exploring factors affecting pedestrians' red-light running behaviors at intersections in China

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ABSTRACT

Pedestrians' Red-light running behavior is one of the most critical factors for pedestrian involved traffic crashes at intersections in China. The primary objective of this study is to explore how various factors affect pedestrians' red-light running behaviors at intersection areas, using the data collected from Hefei, China. A questionnaire was well designed aiming at collecting pedestrians' socio-economic characteristics, trip related features, and attribute variables in different crossing facilities. Based on 631 valid samples, a binomial logistic model was established to evaluate the impacts of contributing factors on pedestrians' red-light running behavior. The modeling results show that four variables significantly affect the probability of pedestrians' red-light running behavior, which are the trip purpose, time period in one day, pedestrian's attitude towards whether to run a red light when in hurry, and pedestrian's attitude towards whether quality of road facility affects crossing behavior. With those variables, the probability of pedestrians' red-light running behavior at intersections could be predicted. Findings of this study can help understand why pedestrians in China run red-lights and identify which pedestrian groups and intersections are more likely to have such behaviors. This study can also help propose countermeasures more efficiently to reduce pedestrian-related crashes at intersections in China.

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1. Introduction

Pedestrian involved traffic crashes has been considered as a serious problem in both traffic safety and public health areas around the world. Over 270,000 pedestrians were killed in traffic crashes each year which accounts for about 22% of total fatalities (World Health Organization, 2014). The safety situation for pedestrians is even worse in developing countries such as China. According to the China Road Accidents Statistics Report, pedestrian involved traffic crashes account for a quarter of all traffic deaths (CRTASR, 2011). A report also shows that there are 127,000 pedestrians running red-lights in city of Shenzhen, China, during the first half year of 2015 (Yi, 2015).

Previously, studies have reported that pedestrians' red-light running behavior is one of the main causes for pedestrian involved traffic crashes at the intersection areas (Liu et al., 2007; Schneider et al., 2010). The red-light running decision of pedestrian is affected by many factors such as personal characteristics, trip attributes, and even pedestrians' attitudes towards red-light running and crossing facilities. It would be interesting to see how those factors affect pedestrians' red-light running behaviors. In addition, though several previous studies have studied pedestrians' red-light running in multiple countries (Florian et al., 2011; Duduta et al., 2014; Medina et al., 2014; Onelcin and Alver, 2015), a study focusing on the topic in China is still necessary in comparing the results between China and other countries.

The primary objective of this study is to explore the factors affecting pedestrians' red-light running behaviors using data collected from a city in China. The study can help better understand why pedestrians run red-lights to propose more effective countermeasures to reduce pedestrian involved traffic crashes at intersections in China. In the following section, a review of relevant studies is provided. In Section 3, the data collection is introduced. In Section 4, the method for analysis is introduced. Section 5 discusses the modeling results. The paper ends with brief concluding remarks in future work in Section 6.

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2. Literature review

In the initial stage, previous studies generally focused on evaluating pedestrians' crossing characteristics. For example, Fruin (1971) examined the spatial characteristics, behavioral characteristics, pedestrian movement patterns to study pedestrian flow. In particular, researchers have analyzed the influential factors of pedestrians' street crossing behaviors from the perspective of individual socioeconomic and psychological characteristics, road environment, etc.

In the recent years, studies have been conducted to evaluate pedestrians' crossing behaviors in some risky situations. Holland and Hill (2007) used the theory of planned behavior (TPB) to explore the effect of gender on the intentions of pedestrians' crossing in a risk situation. They concluded that women were more sensitive to risk perception and less likely to cross street in a high risk situation. Men got a higher safety perception of direct crossings than women (Cantillo et al., 2015). Bernhoft and Carstensen (2008) found that older crossing pedestrians had a higher safety consciousness than younger pedestrians. The older pedestrians always paid more attention to making a safer street-crossing decisions by using safer facilities (zebras, signalized intersection); whereas the younger pedestrians more often focused on how to arrive at the destination more quickly. On the other hand, older pedestrian involved accident occupied a higher proportion of injuries and fatalities because of their fragility, deteriorations of motion perception and judgement ability, and increased reaction time (Li et al., 2003; Snowden and Kavanagh, 2006; Zivotofsky et al., 2012).

Some other studies have examined the pedestrian's crossing speed and the herd mentality effect in the crossing behaviors. The pedestrian crossing speed is an extremely important parameter in operation and safety analysis, and provides important information for intersection geometric and signal design. Previous studies found that in general male pedestrians' walking speed was faster than female (Tanaboriboon and Guyano, 1991; Chandra and Bharti, 2013; Tarawneh, 2001). Also, as the growth of the age, pedestrians' crossing speed would slow down. Griffiths et al. (1984) found that the averaged crossing speed of young pedestrians was 1.72 m/s, 1.47 m/s for the middle-aged, and 1.16 m/s for the elderly. Feng and Wu (2004) observed and analyzed the situation of pedestrian crossing in Harbin. They found that the average pedestrian crossing speed was 1.47 m/s and the crossing speed was 1.74 m/s for 85% of pedestrians. Carey (2005) found that when the pedestrians are crossing in groups, their walking speed tended to be slower.

Different road crossing environment could also lead to different characteristics of pedestrian crossing. Sisiopiku and Akin (2003) studied pedestrians' perceptions in different pedestrian crossing situations such as signalized and unsignalized intersection, marked and non-striped midblock, with and without physical barriers, shelters and colored paving at medians, and pedestrian warning signs. Li (2013) characterized pedestrians' intended waiting time by proposing a U-shaped distributional model which was helpful to understand the mechanism of pedestrians' street-crossing behavior at signalized intersections. The waiting time of pedestrians crossing in Montreal was also found in other studies, which had provided the basis for properly timing pedestrian signals (Brosseau et al., 2013; Yang et al., 2015). In addition, Hamed (2001) found that pedestrians tended not to accept higher risk if they involved in a traffic accident in the past, or they were accompanied by children, or they have heavy luggage, or their mobility is impaired.

Previous studies have considered many methods to study the characteristics of pedestrians' red-light running behaviors. Yagil (2000) proposed a hierarchically multivariate regression models to examine pedestrians self-reported road-crossing behavior in relation to beliefs regarding the consequences of the behavior incorporated in the health belief model, instrumental and normative motives for compliance with safety rules, and situational factors. Diaz (2002) developed a model of structural equation which consisted of attitude, subjective norm, perceived control, behavioral intention and reported violations, errors and lapses. Li (2014) established a bi-level multivariate model to analyze the risk of pedestrians' unsafe street crossing behavior at signalized intersections. Cantillo et al. (2015) proposed a hybrid framework to analyze the pedestrians' crossing choice on the urban road. Through investigating and analyzing, we found that crossing the street with a minor could reduce risk.

Some literatures on the topic of characteristics of pedestrian crossing were also found. Along with the widely use of mobile telephone, the man who used mobile while road crossing was more likely to cause traffic accidents. Most conclusions suggested that talking on a mobile phone is associated with cognitive distraction that may undermine pedestrian safety (Bungum et al., 2005). A research showed that females appeared to pay less attention to traffic both before and during crossing and a similar pattern of results for males at unsignalized crossings may have reached significance in a larger sample. Further, talking on a mobile was associated with slower crossing speeds for females at signalized crossings, and for males at unsignalized crossings. These effects suggested that talking on a mobile phone is associated with cognitive distraction that may undermine pedestrian safety (Hatfield and Murphy, 2007).

In summary, previous studies have evaluated why pedestrians run a red-light considering individual socioeconomic features, psychological characteristics, and road environment factors. However, previous studies seldom considered the impacts of pedestrians' attitude related factors on the red-light running behaviors. As human behaviors are affected by attitudes, it would be necessary to consider such factors in the analysis. Furthermore, traffic environment, cultural background, and travelers' decisions making process in China are quite different from those in other countries. Thus, it is very necessary to conduct a research on exploring what factors affect pedestrians' red-light running behaviors at intersections in China.

3. Data

3.1. Survey design

In order to capture the potential influencing factors on pedestrian's red-light running decision, a SP survey which includes fifteen intelligible items is designed to obtain the individual characteristics, trip characteristics and pedestrian's attitude. The survey collects three parts of information. The first part addresses questions on pedestrian individual socioeconomic characteristics such as gender (coded 0 = female, 1 = male), age (coded 0 = less than 12, 1 = 12-18, 2 = 19-25, 3 = 26-35, 4 = 36-55, 5 = more than 55), education level (coded 0 = primary school, 1 = junior middle school, 2 = senior middle school, 3 = junior college, 4 = undergraduate, 5 = postgraduate), profession (coded 0 = student, 1 = worker, 2 = attending staff, 3 = civil servant, 4 = individual worker, 5 = retiree, 6 = others) and income (coded 0 = nearly no, 1 = less than 10000, 2 = 10000-30000, 3 = 30000-50000, 4 = 50000-70000, 5 = 70000-100000, 6 = more than 100000).

The second part addresses questions on trip characteristics such as trip purpose, time requirement, intersection familiarity, the tolerable waiting time and the time segment that running a red-light more easily. Trip purpose was divided into 5 groups (coded 0 = go work or go to school, 1 = living include entertainment, shopping, see a doctor etc., 2 = stroll, 3 = go home and 4 = others). Time requirement was divided into 2 groups (coded 0 = harsh and 1 = not harsh). Intersection familiarity was divided into 3 groups (coded 0 = familiar, 1 = general and 2 = unfamiliar). Tolerable waiting time

(unit: second) was divided into 5 groups (coded 0 = less than 40, 1 = 41-60, 2 = 61-90, 3 = 91-99, 4 = more than 99). Travelers' trip purpose and psychology are very different at different time periods in a whole day, and thus may affect the crossing behaviors at intersections. In this study, four different time segments are considered which are dawn (coded 0 = from 3 to 5 am), daytime (coded 1 = from 5 am to 5 pm), dusk (coded 2 = from 5 pm to 7 pm), and night (coded 3 = from 7 pm to 3 am).

The third part addresses questions on pedestrian's attitude in different crossing environment such as: whether the presence of voice remind will affect your crossing behavior, whether you will run a red-light when you are in hurry, whether you will run a red-light during a small traffic volume, whether you will wait the red signal during trip with elders or children and whether the imperfect of road facilities will affect your crossing behavior. All of questions were divided into 2 groups (coded 0 = yes and 1 = no).

3.2. Data collection

The survey was conducted in Hefei, China. Hefei is the capital of Anhui province and is one typical large-sized city that locates in the east of China. By the end of 2014, the population was 7,696,000 and the area was 11,408.48 km^2. In recent years, the economy of Hefei has realized fast development, and the GDP reached 515.8 billion yuan in 2014, increased by over 10% last year. The daily trip time each person in Hefei was about 3, the main trip mode were walk, electric vehicle and bus, the ratio for them were26.0, 27.4, 18.4, respectively. The geometric design, traffic flow state and control feathers of intersections in Hefei can represent the most common situations in major cities of China.

To ensure the universality and abundance of sampling, the following criteria were applied when selecting the intersections for study: (1) intersection is signalized with pedestrian signal lamp; (2) intersection is closed to the central business district or transit stations so that the pedestrian flow rate is large; and (3) intersection without presence of police so that red-light running behavior can be observed. After the selection, six typical signalized intersections were carefully selected to conduct the investigation.

The field investigation was carried out by nine research assistants in the school, holding a survey card issued by the traffic management department of the local transportation government. The investigation time was peak period (from 7:30 to 8:30 a.m.; from 11:30 a.m. to 12:30 p.m.; from 6:00 to 7:00 p.m.) in both weekdays and weekends during May to June, 2015. During the investigation, pedestrians surveyed were asked to fill out a pedestrian crossing street questionnaire which takes about five minutes to answer. The meaning of each term was explained to the target people to ensure the veracity of the investigation. Finally, 650 responses were received. Later, questionnaires that have multiple choices or no choice were excluded from analysis. At last, a total of 631 valid samples were obtained.

The personal socioeconomic characteristics of the surveyed sample are summarized in Table 1. Of the 631 surveyed pedestrians, 54.5% are male. Most of the pedestrians are in the groups aged 19–25 years, which accounts for 49.7%. Within the sample, most of pedestrians' education levels are junior college and undergraduate, which accounts for 61%. In the sample, student accounts for 32% and almost 35% pedestrians' income is nearly zero. We could also get that the red-light running percent is very great especially in the groups aged 19–25 years, junior college education level, student group and annual income of 30000–50000, which the proportions are 84.9%, 82.8%,81.2% and 82.1% respectively.

The trip characteristics of the surveyed samples are summarized in Table 2. The pedestrians' trip purpose mainly centralized on going to working or school, daily life such as entertainment, shopping and so on. More than 50% pedestrians stated that trip time requirement is not very stringent, but there were still 365 pedestrians running a red-light, which accounted for 74.6%. Only 96 pedestrians stated that they were unfamiliar with intersection. As the increasing of tolerable waiting time, the percent of red-light running decreased and there was 46.9% pedestrians running a red-light when the tolerable waiting time was more than 99 s. More than 250 pedestrians held that night was the time segment that running a red-light more easily.

The characteristics of pedestrian's attitude in different crossing environment are summarized in Table 3. Of the 631 surveyed pedestrians, 37.7% pedestrians said that lack of voice remind would affect their crossing behavior and nearly 95% pedestrians run a red-light in this group. Statistic showed that pedestrians would run a red-light more easily when they were in hurry, which accounted for 87.1%. Most of pedestrians were willing to wait the red signal during a small traffic volume or tripping with elders or children, which the frequencies were 433 and 489 respectively. 398 pedestrians held that the perfect degree of road facilities would also affect pedestrians' crossing behavior and nearly 85% pedestrians ran a red-light in this group.

4. Methodology

In our analysis, the response variable for modeling pedestrians' red-light running behavior is binary: run a red-light and not run a red-light. Thus, the binominal logistic model was selected in this study. The binominal logistic model is developed to predict a binary dependent variable as a function of a series of predicting variables, and has been widely used in traffic safety studies (Valent et al., 2002; Simoncic, 2001; Tay et al., 2008). In the model, Y = 1 denotes running a red-light and Y = 0 denotes running during green light. The model form is given by:

$$\log it\left(\frac{p_i}{1-p_i}\right) = b + \alpha_1 x_1 + \alpha_2 x_2 \cdots + \alpha_i x_i \cdots + \alpha_n x_n \tag{1}$$

where p_i is the probability that events occurred, which is the function of a vector of independent variable x_i where a_i is the corresponding coefficient of x_i and b is the intercept of the equation.

Then the condition probability of events occurred is given in Eq. (2).

$$\begin{aligned} p_i &= p(y_i = 1|x_1, x_2, \cdots, x_n) \\ &= \frac{\exp(b + \alpha_1 x_1 + \alpha_2 x_2 \cdots + \alpha_i x_i \cdots + \alpha_n x_n)}{1 + \exp(b + \alpha_1 x_1 + \alpha_2 x_2 \cdots + \alpha_i x_i \cdots + \alpha_n x_n)} \end{aligned} \tag{2}$$

The Score test was used to validate whether the possible contributory factors had a statistically significant relationship with pedestrians' crossing behavior. The Score test procedure was used to estimate the intimate level between independent variable and dependent variable by calculating the values of Score and p (Lekdee and Ingsrisawang, 2010; Chen and Chen, 2013). The variables were included in the model step by step and those that were not significant at a 95% confidence level were excluded from the model. The variable selection procedure was repeated multiple times while monitoring the model statistical performance.

Fifteen explanatory variables were considered in this model. For the explanatory variables, three explanatory variables are continuous variables, which are age, annual income and the tolerable waiting time, and the others are categorical variables. Some of the categorical variables had more than two levels, the creation of dummy variables was necessary to represent the data and match above mentioned model in this paper. One way to code the categorical variables is to have k-1 design variables for the k levels of that variable.

An example of this method is given in Table 4 for the variable of pedestrians' profession (PRO for short), which has seven levels,

Table 1
Pedestrian individual socioeconomic characteristics.

Attribute	Range	Frequency	Percent	Frequency of running red-light	Percent of running red-light
gender	female	287	45.5%	210	73.2%
	male	344	54.5%	259	75.3%
age	less than 12	4	0.6%	2	50.0%
	12–18	49	7.8%	31	63.3%
	19–25	311	49.3%	264	84.9%
	26–35	149	23.6%	112	75.2%
	36–55	87	13.8%	45	51.7%
	more than 55	31	4.9%	15	48.4%
education level	primary school	20	3.2%	9	45.0%
	junior middle school	96	15.2%	61	63.5%
	senior middle school	116	18.4%	80	69.0%
	junior college	209	33.1%	173	82.8%
	undergraduate	176	27.9%	135	76.7%
	postgraduate	14	2.2%	11	78.6%
profession	student	202	32.0%	164	81.2%
	workers	51	8.1%	36	70.6%
	attending staff	44	7.0%	32	72.7%
	civil servant	116	18.4%	88	75.9%
	individual worker	134	21.2%	100	74.6%
	retiree	16	2.5%	8	50.0%
	others	68	10.8%	41	60.3%
annual income (Rmb)	nearly no	218	34.5%	164	75.2%
	less than 10000	75	11.9%	54	72.0%
	10000–30000	115	18.2%	82	71.3%
	30000–50000	112	17.7%	92	82.1%
	50000–70000	56	8.9%	37	66.1%
	70000–100000	35	5.5%	26	74.3%
	more than 100000	20	3.2%	14	70.0%

Table 2
Trip characteristics of the surveyed samples.

Attribute	Range	Frequency	Percent	Frequency of running red-light	Percent of running red-light
trip purpose	work or go to school	196	31.1%	154	78.6%
	living (entertainment, shopping, etc.)	304	48.2%	230	75.7%
	stroll	52	8.2%	32	61.5%
	go home	34	5.4%	25	73.5%
	others	45	7.1%	28	62.2%
time requirement	harsh	142	22.5%	104	73.2%
	not harsh	489	77.5%	365	74.6%
intersection familiarity	acquaintance	334	52.9%	231	69.2%
	general	201	31.9%	166	82.6%
	unacquaintance	96	15.2%	72	75.0%
the tolerable waiting time (second)	less than 40	180	28.5%	156	86.7%
	41–60	230	36.5%	182	79.1%
	61–90	87	13.8%	61	70.1%
	91–99	53	8.4%	32	60.4%
	more than 99	81	12.8%	38	46.9%
the time segment that running a red-light more easily	dawn	69	10.9%	41	59.4%
	daytime	207	32.8%	167	80.7%
	dusk	102	16.2%	75	73.5%
	night	253	40.1%	186	73.5%

Table 3
The characteristics of pedestrian's attitude in different crossing environment.

Attribute	Range	Frequency	Percent	Frequency of running red-light	Percent of running red-light
Whether the presence of voice remind will affect your crossing behavior	yes	238	37.7%	225	94.5%
	no	393	62.3%	244	62.1%
Whether you will run a red-light when you are in hurry	yes	342	54.2%	298	87.1%
	no	289	45.8%	171	59.2%
Whether you will wait the red signal during a small traffic volume	yes	433	68.6%	306	70.7%
	no	198	31.4%	163	82.3%
Whether you will wait the red signal during trip with elders or children	yes	489	77.5%	359	73.4%
	no	142	22.5%	110	77.5%
Whether the imperfect of road facilities will affect your crossing behavior	yes	398	63.1%	328	82.4%
	no	233	36.9%	141	60.5%

and hence six design variables are needed. When the pedestrian is a student, the six design variables: PRO1, PRO2, PRO3, PRO4, PRO5 and PRO6 would all be set to equal 0. When the pedestrian is a worker, PRO1 would be set equal to 1, whereas PRO2, PRO3, PRO4, PRO5 and PRO6 would still equal 0, and so forth for the other respondents.

Table 4
Example of coding profession variable.

Profession	Parameter coding					
	PRO1	PRO2	PRO3	PRO4	PRO5	PRO6
student	0	0	0	0	0	0
worker	1	0	0	0	0	0
attending staff	0	1	0	0	0	0
civil servant	0	0	1	0	0	0
Individual worker	0	0	0	1	0	0
retiree	0	0	0	0	1	0
others	0	0	0	0	0	1

5. Results

5.1. Model estimation

The binomial logistic model of pedestrians' crossing behavior which based on SP survey data was constructed using the aforementioned methodology. The statistical software (SPSS 19.0) was used to implement this process. Table 5 lists the results of score-test based on 0.05 significant levels.

The score-test results in Table 5 show that thirteen independent variables were found to be significantly at a 95% confidence level related to pedestrians' red-light running behaviors, except for two variables which are the gender and the attitude variable whether pedestrians would wait the red signal during tripping with elders or children. Then those two insignificant variables were rejected and other significant variables were considered in the binomial logit model to do the deep interactive calculation. The modeling results are given in Table 6 through iterative calculation. At last, four independent variables were found to be significant in model estimates, which are the trip purpose, time period in a day, the attitude variable whether to run a red light when in hurry, and the attitude variable whether imperfect road facility affect crossing behavior. The probability of pedestrains' red-light running behavior can be calculated based on results in Table 6. The probability prediction function is given by:

$$p = \frac{\exp(1.383 - 0.932TRP2 - 1.206INH + 1.202TIS1 + 0.830TIS2 + 0.732TIS3 - 0.728IMR)}{1 + \exp(1.383 - 0.932TRP2 - 1.206INH + 1.202TIS1 + 0.830TIS2 + 0.732TIS3 - 0.728IMR)} \quad (3)$$

5.2. Discussion

5.2.1. Trip characteristics

Four trip purposes were considered in the modeling with the purpose of going to work/school as the reference category. The results in Table 6 show that the effect of stroll trip purpose has a significant impact on pedestrian's red-light running behaviors at a 95% confidence level. The odds ratio is $e^{-0.932} = 0.394$. This means that the odds ratio of the red-light running probability of strolling group is just 0.394 times of those office-goers or students. The coefficient of the variable is negative, suggesting that strolling people should be more likely to have a safe crossing behavior as compared with those who are going to work/school. Pedestrians whose purpose is going to work or school usually want to short the consuming time to arrive at their destination so that they are more likely to cross street ignoring traffic regulations (Zou et al., 2012). The impacts of other trip purposes are not significant probably because those people are also in hurry during those trips. The red-light running probabilities of these groups are similar to the reference group whose street-crossing behaviors have a high risk. More attention should be paid to the traffic regulations and accidents propaganda to the pedestrian groups with those trip purposes. More attention should be paid to the traffic regulations and accidents propaganda to the pedestrian groups with those trip purposes. In addition, transportation agencies may also propose policies to punish pedestrian's red-light running behaviors at intersections, especially in rush hours. Those policies could be fine pedestrians who run a red-light and conduct on-spot educations. Another potential way is to link pedestrians' red-light running behaviors with their individual credits.

The time period that pedestrian running a red-light is modeled as three dummy variables with the dawn considered as the reference level. Table 6 shows that there is significant different between the time period variables on pedestrians' crossing behavior. The coefficients of these variables are all positive which means that the red-light running probability during dawn period is the smallest. The red-light running probability of daytime variable is the largest, which means pedestrians who crossing street during the daytime is more likely to run red-light as compared to other time segment. The reason is that people's daily activities such as working, shopping etc. are mainly conducted during the daytime so that the crossing pedestrian flows are higher. The crossing constitution of pedestrians is more complex and the safety consciousness is more irregularity during day time. Hence, controls for pedestrians should be strengthened especially in peak hours to guarantee pedestrians' safety and heighten traffic efficiency in Hefei city.

The red-light running probability in dusk and night segment is also found to be higher than that in dawn which is a little counterintuitive. The reason is that unlike some other countries, there are a lot of activities of night life in Hefei and other cities in China. People choose to go shopping and do some entertainments during dusk and night time, resulting in increasing crossing activities at intersection areas. Considering the reduced visibility, sight distance and impact of glare in night, more crashes are likely to occur at intersections due to the red-light crossing behaviors by pedestrians. Perceptions of control in potentially dangerous road-crossing situations will be beneficial in increasing pedestrians' risk awareness (Liu and Tung, 2014; Evans and Norman, 1998; Ajzen, 1991). A control strategy of variable speed limits considering location and time determination should be developed to endure pedestrians' street-crossing safety. Meanwhile, the sufficient lighting of intersections in dusk and night segment is also absolutely necessary. Strategies such as dynamic speed limit control considering location and time determination could be proposed to reduce vehicle speeds to improve pedestrian crossing safety. In addition, providing sufficient lighting at intersections in dusk and night is another effective way to improve pedestrian crossing safety.

5.2.2. Attitude variables

Pedestrians' attitudes were also found to be significantly related to the red-light running behaviors. According to the modeling results in Table 6, pedestrians' attitude towards whether to run a red light when in hurry is a significant factor. The negative coefficient of the variable indicates that pedestrians in hurry are more likely to run a red light which is quite straightforward. The odds ratio is $e^{-1.206} = 0.299$. This means that the odds ratio of the red-light running probability of group who is not in hurry is 0.299 times of susceptible group. One possible reason would be that when the waiting time of pedestrians affected by a hurry exceeds the tolerable limit, they will generate exigent psychology, and then, run a red-light neglecting crossing safety. And these results are in line with previous studies showing that saving time and convenience are the major reasons of pedestrians' red-light running behaviors (Ren et al., 2011). Meanwhile, other pedestrians in hurry could follow them to run a red-light due to the effect of herd mentality,

Table 5
Score test analysis of independent variables.

Independent variables	Abbreviation	Score	df	Sig.
Gender	GEN	0.369	1	0.544
Age (base: <12 years old)		56.837	5	0.000
12–18 years old	AGE1	3.406	1	0.065
19–25 years old	AGE2	35.844	1	0.000
26–35 years old	AGE3	0.072	1	0.788
36–55 years old	AGE4	27.016	1	0.000
>55 years old	AGE5	11.496	1	0.001
Education level (base: primary school)		25.084	5	0.000
Junior middle school	EDL1	6.901	1	0.009
Senior middle school	EDL2	2.141	1	0.143
Junior college	EDL3	11.690	1	0.001
Undergraduate	EDL4	0.723	1	0.395
Postgraduate	EDL5	0.135	1	0.713
Profession (base: student)		17.545	6	0.007
Worker	PRO1	0.406	1	0.524
Attending staff	PRO2	0.063	1	0.801
Civil servant	PRO 3	0.176	1	0.675
Individual worker	PRO 4	0.008	1	0.929
Retiree	PRO 5	5.091	1	0.024
Others	PRO 5	7.864	1	0.005
Annual income (base: nearly no. unit: rmb)		6.638	6	0.356
<10000	ANI1	0.241	1	0.623
10000–30000	ANI2	0.673	1	0.412
30000–50000	ANI3	4.360	1	0.037
50000–70000	ANI4	2.195	1	0.138
70000–100000	ANI5	0	1	0.995
>100000	ANI6	0.203	1	0.653
Trip purpose (base: go work or go to school)		10.056	4	0.039
Living (shopping, see a doctor etc.)	TRP1	0.545	1	0.460
Stroll	TRP2	4.857	1	0.028
Go home	TRP3	0.012	1	0.913
Others	TRP4	3.720	1	0.054
Time requirement	TIR	0.113	1	0.736
Intersection familiarity (base: acquaintance)		11.88	2	0.003
General	INF1	10548	1	0.001
Unacquaintance	INF2	0.027	1	0.870
The tolerable waiting time (base: <40. unit: second)		31.258	4	0.000
41–60	TOW1	1.782	1	0.182
61–90	TOW2	0.008	1	0.929
91–99	TOW3	0.734	1	0.392
>99	TOW4	30.300	1	0.000
The time period in a day that running a red-light easier (base: dawn)		12.529	3	0.006
Daytime	TIS1	6.509	1	0.011
Dusk	TIS2	3.041	1	0.030
Night	TIS3	4.1450	1	0.046
Whether the presence of voice remind will affect your crossing behavior	PRV	24.089	1	0.000
Whether you will run a red-light when you are in hurry	INH	64.194	1	0.000
Whether you will run a red-light during a small traffic volume	SMT	9.670	1	0.002
Whether you will wait the red signal during tripping with elders or children	ELC	0.946	1	0.331
Whether the imperfect of road facilities will affect your crossing behavior	IMR	36.928	1	0.000

which is consistent with past studies (Yagil, 2000; Zhou et al., 2009). So we should strengthen law enforcement, propaganda and education of the harmful influence of running red light in hurry especially in the area with a faster life rhythm, such as central business district. In addition, electronic facilities are suggested to be installed densely in this area to remind pedestrians to observe traffic regulations.

Pedestrian's attitude towards whether quality of road facility (such as pedestrians' signal lamp, signal timing, zebra crossing, and monitor facilities) affects crossing behavior was found to be a significant factor in the model estimates. The variable has a negative coefficient indicating that pedestrians affected by the road facility quality are more likely to run a red light at intersections. The odds ratio is $e^{-0.728} = 0.483$ suggesting that the red-light running probability of the pedestrians who is unaffected is 0.483 times of those affected. Zebra crossing and pedestrians' signal lamp can be used to standardize pedestrians' crossing behavior and improve pedestrians' crossing safety. At those intersections without monitoring equipment, pedestrians are more likely to run a red light because their behaviors will not be recorded. These findings are also consistent with previous studies which reported that perfect road facilities such as the presence of a pedestrian signal can decreases significantly the probability of dangerous crossings and violations (Lavalette et al., 2009; Francisoco, 2006; Brosseau et al., 2013). So improvement of road crossing facilities and traffic signal control system could help to reduce the red-light running behaviors and create a safer crossing environment for pedestrians.

Table 6
Parameter estimates of the model.

Variables	Coefficient	Standard error	P	Odds ratio	95% Confidence interval	
					Lower	Upper
TRP2	−0.932	0.398	0.019	0.394	0.181	0.859
TIS1	1.202	0.360	0.001	3.326	1.643	6.730
TIS2	0.830	0.396	0.036	2.294	1.056	4.983
TIS3	0.723	0.337	0.032	2.060	1.065	3.985
INH	−1.206	0.227	0.000	0.299	0.192	0.467
IMR	−0.728	0.221	0.001	0.483	0.313	0.745
Constant	1.383	1.717	0.421	3.986		

6. Conclusions

The study explored the factors affecting pedestrians' red-light running behaviors in Hefei, China. Except for individual characteristics, trip related features and the pedestrians' attitudes were also considered as explanatory variables in the analysis. Field survey was conducted and binary logistic regression model was used to estimate the red-light running decision by pedestrians. More specifically, the following results were obtained: On the aspect of trip characteristics, (a) strolling people are more likely to have a safe crossing behavior, (b) pedestrians' red-light running probability during daytime is the largest and they are also more likely to run a red-light in dusk and night than that in dawn. On the aspect of pedestrian's attitude towards red-light running behaviors, (a) pedestrians affected by the road facility quality are more likely to run a red light at intersections, (b) pedestrians in hurry are more likely to run a red light.

In our study, due to the limitation of time and labor resources, data were collected in one Chinese city. The results may not be exactly the same in other cities or countries due to the difference in culture, driving behaviors, etc. However, the experiment design, modeling framework, and results are helpful for researchers in other cities or countries to do similar analyses. In other Chinese cities, as the cultural background and driving styles are similar as in Hefei, our results may hold for other Chinese cities and will be helpful in understanding red-light running behaviors. In some other countries, the factors related to red-light running behaviors would be different. It would be interesting to compare their results with our study and identify how cultural and driving factors affect the difference in red-light running behaviors in different countries.

Though the sample size in our study is sufficient for obtaining a reasonable accurate model performance, more data should be collected at multiple intersections in other cities to validate the findings of our study. The same procedure as proposed in this study can be followed to obtain the impacts of contributing factors. In addition, this study only considered pedestrians' red-light running behaviors at signalized intersection areas. In urban areas there are several other types of crossing facilities such as those located within street segments with/without crosswalk buttons. It would be interesting to identify how pedestrians' red-light running behaviors are affected by various factors and compare the results with signalized intersections. Furthermore, the current study focused on individual's crossing behaviors while ignoring the impacts of pedestrian platoons. Pedestrians' crossing behaviors could be affected by the behaviors of other people and study is required to evaluate such effects. The authors recommend that future studies may focus on those issues.

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